



Student Instructions:

- Write your complete ID code on your answer sheet.
 - If you change your answer, erase well.
1. Three of the following materials are good insulators of heat and electricity. Which one is a conductor?
 - A. wood
 - B. rubber
 - C. plastic
 - D. steel
 2. Which choice below is a physical property of matter?
 - A. acidity
 - B. volume
 - C. toxicity
 - D. flammability
 3. Which one of the following processes occurs when a substance transforms directly from a solid to a gas without passing through a liquid phase?
 - A. evaporation
 - B. condensation
 - C. sublimation
 - D. precipitation
 4. Below are four simple machines. Which one is made from an inclined plane wrapped around a cylinder?
 - A. wheel and axle
 - B. pulley
 - C. screw
 - D. wedge
 5. Newton's First Law of Motion concerns it. Which of the following words names an object at rest's tendency to remain at rest, or an object in motion's tendency to remain in motion, unless acted on by an outside force?
 - A. inertia
 - B. thrust
 - C. drag
 - D. friction
 6. Below are four parts of an electrical circuit. Which one's primary function is turning the circuit on or off?
 - A. load
 - B. conductor
 - C. power source
 - D. switch
 7. Which one of the following metals is naturally magnetic?
 - A. gold
 - B. iron
 - C. silver
 - D. copper

8. Which of these rocks is metamorphic?
- A. limestone
 - B. basalt
 - C. slate
 - D. shale
9. Because it retains nutrients and moisture, it is good for farming. Which choice below names this rich, crumbly soil that also contains organic matter called humus?
- A. loam
 - B. silt
 - C. sand
 - D. clay
10. Match each step in the rock cycle with its description.
- | | |
|---------------|---|
| x. weathering | 1. brings underground rocks to Earth's surface |
| y. uplift | 2. gravity, wind, water, or ice moves soil and rock particles |
| z. erosion | 3. breaks bedrock into sediment |
- A. x-2, y-1, z-3
 - B. x-2, y-3, z-1
 - C. x-3, y-2, z-1
 - D. x-3, y-1, z-2
11. The temperature difference can be small or large, depending on the nature of the two air masses. Which of the following weather fronts moves extremely slowly, if it moves at all?
- A. warm front
 - B. cold front
 - C. stationary front
 - D. occluded front
12. Below are four atmospheric layers. Which one, the lowest, contains most of Earth's clouds and weather systems?
- A. exosphere
 - B. mesosphere
 - C. thermosphere
 - D. troposphere
13. Which one of the following astronomical objects is small and rocky and, in most cases, exists in a "belt" between Mars and Jupiter?
- A. asteroid
 - B. comet
 - C. meteor
 - D. star
14. Which two planets in our solar system have no known moons?
- A. Uranus and Neptune
 - B. Jupiter and Saturn
 - C. Mercury and Venus
 - D. Earth and Mars

15. Many plant cells contain chloroplasts. Plant-like members of which kingdom also have chloroplasts?
- A. Bacteria
 - B. Protista
 - C. Fungi
 - D. Animalia
16. Some protists move using flagella, cilia, or pseudopods. Which one of the protists below does NOT move using one of these appendages?
- A. dinoflagellate
 - B. amoeba
 - C. paramecium
 - D. diatom
17. Eukaryotic cells contain thousands of them. Which of the following organelles is the “powerhouse of the cell” because it generates energy?
- A. ribosome
 - B. vacuole
 - C. lysosome
 - D. mitochondrion
18. Which choice below is NOT one of the functions of a plant’s roots?
- A. photosynthesizing
 - B. absorbing water
 - C. anchoring the plant in the soil
 - D. storing food
19. Three of the following plants are vascular. Which one is non-vascular?
- A. moss
 - B. corn
 - C. cactus
 - D. fern
20. Match each invertebrate with its phylum.
- | | |
|--------------|--------------------|
| x. earthworm | 1. Platyhelminthes |
| y. tapeworm | 2. Annelida |
| z. sea star | 3. Echinodermata |
- A. x-3, y-2, z-1
 - B. x-2, y-1, z-3
 - C. x-1, y-2, z-3
 - D. x-1, y-3, z-2
21. Which one of the following words can be used for an adult insect?
- A. nymph
 - B. imago
 - C. larva
 - D. pupa

22. Three of the animals below are warm-blooded. Which one is cold-blooded?
- A. rattlesnake
 - B. husky
 - C. pot-bellied pig
 - D. rooster
23. Which one of the following animals is an herbivore?
- A. wolf
 - B. bear
 - C. bald eagle
 - D. deer
24. Bacteria, viruses, or fungi can cause it. Which condition below causes lung tissue inflammation, making breathing difficult?
- A. psoriasis
 - B. arthritis
 - C. pneumonia
 - D. chicken pox
25. Below are four parts of the human skeletal system. Which one is made of 33 vertebrae divided into cervical, thoracic, lumbar, sacral, and coccyx regions?
- A. spine
 - B. humerus
 - C. femur
 - D. ribs
26. Which of the following steps comes LATEST in the scientific method?
- A. Ask a question.
 - B. Communicate your results.
 - C. Construct a hypothesis.
 - D. Conduct an experiment.
27. They can hold liquids for heating or mixing, as well as grow bacteria. What piece of laboratory equipment is pictured here?
- A. dropper
 - B. beaker
 - C. crucible
 - D. test tube
28. An entomologist is a scientist who studies which of the following?
- A. fish
 - B. birds
 - C. insects
 - D. reptiles
29. He studied it in hopes of helping poor cotton farmers rejuvenate their soil. George Washington Carver devised over 300 uses for which plant below?
- A. peanut
 - B. apple
 - C. corn
 - D. wheat



30. Match each scientist with his invention.

- x. Isaac Newton
- y. Thomas Edison
- z. Benjamin Franklin

- 1. a type of reflecting telescope
- 2. bifocals
- 3. phonograph

- A. x-3, y-1, z-2
- B. x-1, y-2, z-3
- C. x-1, y-3, z-2
- D. x-2, y-3, z-1

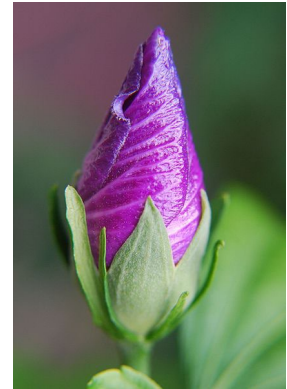
TIEBREAKERS 31-36

31. It is a layer of ionized gas. Which choice below names the sun's visible surface?

- A. supernova
- B. photosphere
- C. quasar
- D. aphelion

32. In this photo, the green, leaf-like structures that protect the budding flower are called what?

- A. petals
- B. pistils
- C. sepals
- D. stamens



33. It is the largest marine phylum and the second-largest invertebrate phylum. Clams, snails, and octopi are all part of which phylum?

- A. Porifera
- B. Mollusca
- C. Arthropoda
- D. Cnidaria

34. Estivation is the process of spending a hot, dry season in a dormant state. Which one of the following animal species estivates?

- A. desert tortoise
- B. brown bat
- C. groundhog
- D. red squirrel

35. Radiation from which one of the following electromagnetic spectrum categories has the longest wavelengths?

- A. gamma rays
- B. ultraviolet
- C. infrared
- D. radio

36. Louis Pasteur did NOT accomplish which one of the following?

- A. developed an anthrax vaccine
- B. discovered that humans inhale oxygen
- C. developed a rabies vaccine
- D. discovered that microbes cause fermentation

**Science – Elementary – District
2024 Governor’s Cup**

1. D
2. B
3. C
4. C
5. A
6. D
7. B
8. C
9. A
10. D
11. C
12. D
13. A
14. C
15. B
16. D
17. D
18. A
19. A

20. B
21. B
22. A
23. D
24. C
25. A
26. B
27. D
28. C
29. A
30. C
31. B
32. C
33. B
34. A
35. D
36. B

TIEBREAKERS 31-36

Notes for Inquiry Panel

[illegible]

1: 1, 2, 3
2: 4, 5
3: 6, 7, 35
4: 8, 9, 10
5: 11, 12
6: 13, 14, 31
7: 15, 16, 17
8: 18, 19, 32
9: 20, 21, 33
10: 22, 23, 34
11: 24, 25
12: 26, 27, 28
13: 29, 30, 36